Eliminate Rabies Trial Analysis Report

Eliminating human rabies: impact of enhanced vaccination coverage (main coverage definition)

Report description:
Final Report Version 04

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R version 4.5.1 (2025-06-13) for Unix with packages lubridate, foreign, Hmisc, gdata, lme4, glmmTMB, DHARMa, ggplot2, sjPlot, parallel, forcats, brms, bayesplot, coda, see and vioplot

Table 1. Characteristics of wards, overall and by trial arm.

		Total (n=112)	Trial Arm	
			Team-based (N=56)	Community-based (N=56)
District	Bunda	15 (13%)	7 (12%)	8 (14%)
	Butiama	14 (12%)	7 (12%)	7 (12%)
	Musoma	28 (25%)	15 (27%)	13 (23%)
	Rorya	17 (15%)	8 (14%)	9 (16%)
	Serengeti	13 (12%)	7 (12%)	6 (11%)
	Tarime	25 (22%)	12 (21%)	13 (23%)
Land use	Rural	83 (74%)	37 (66%)	46 (82%)
	Semi-Urban	7 (6%)	5 (9%)	2 (4%)
	Urban	22 (20%)	14 (25%)	8 (14%)
N dogs surveyed	Mean (SD)	164 (55)	165 (49)	163 (61)
	Median (IQR)	162 (122, 192)	171 (122, 190)	159 (122, 192)
	[Range]	[49, 444]	[76, 301]	[49, 444]
N dogs (estimated)	Mean (SD)	1814 (802)	1762 (847)	1866 (758)
	Median (IQR)	1784 (1329, 2177)	1681 (1233, 2177)	1838 (1516, 2162)
	[Range]	[100, 4891]	[236, 4891]	[100, 4071]
N vaccinations	Mean (SD)	1669 (961)	1050 (484)	2289 (920)
	Median (IQR)	1584 (957, 2198)	1002 (676, 1305)	2217 (1756, 2684)
	[Range]	[139, 5160]	[167, 2096]	[139, 5160]

Table 2. Primary analysis and secondary analysis 1. Coverage and intervention odds ratio estimates (95% CI) at each survey estimated from the GLMM reported in the previous table. P-values at each survey are from likelihood ratio tests of the primary analysis null hypothesis of no intervention effect at any of the six surveys and secondary analysis 1 null hypothesis of no intervention effect at any of the three visit 2 surveys. Mean coverage (95% CI), odds ratio (95% CI) and P-value across all years were estimated from 10000 parametric bootstrap samples. Population: all dogs with primary outcome data.

Year-visit	Coverage (95% CI)		Odds ratio (95% CI)	P-value	P-value (primary analysis)	P-value (secondary analysis 1)
	Team	Community				
Y1-V1	0.32 (0.25, 0.41)	0.53 (0.44, 0.62)	2.36 (1.74, 3.20)	<0.001	<0.001	<0.001
Y1-V2	0.22 (0.16, 0.29)	0.49 (0.40, 0.58)	3.49 (2.53, 4.81)	<0.001		
Y2-V1	0.40 (0.32, 0.49)	0.50 (0.41, 0.58)	1.48 (1.10, 2.01)	0.011		
Y2-V2	0.31 (0.24, 0.40)	0.57 (0.48, 0.65)	2.90 (2.13, 3.93)	<0.001		
Y3-V1	0.46 (0.37, 0.54)	0.62 (0.53, 0.70)	1.96 (1.45, 2.66)	<0.001		
Y3-V2	0.31 (0.24, 0.39)	0.54 (0.45, 0.63)	2.62 (1.93, 3.56)	<0.001		
All years	0.37 (0.28, 0.46)	0.55 (0.45, 0.65)	2.09 (1.58, 2.75)	<0.001		

Table 3. Secondary analysis 2. Estimated probability (95% CI) of village-level coverage below 40% by month and averaged over the trial year, for each trial arm. The distribution of coverage among villages was estimated from 10000 parametric bootstrap samples. Population: all dogs with primary outcome data.

Trial month	P(Coverage	
	Team-based	Community-based
1	0.40 (0.19, 0.62)	0.20 (0.05, 0.41)
2	0.43 (0.21, 0.66)	0.19 (0.05, 0.40)
3	0.46 (0.23, 0.70)	0.19 (0.04, 0.40)
4	0.50 (0.26, 0.74)	0.18 (0.04, 0.40)
5	0.54 (0.30, 0.78)	0.17 (0.03, 0.40)
6	0.59 (0.33, 0.82)	0.17 (0.02, 0.40)
7	0.63 (0.38, 0.86)	0.17 (0.02, 0.40)
8	0.67 (0.42, 0.89)	0.17 (0.02, 0.40)
9	0.71 (0.46, 0.91)	0.18 (0.02, 0.41)
10	0.75 (0.51, 0.93)	0.18 (0.03, 0.42)
11	0.77 (0.54, 0.94)	0.20 (0.04, 0.43)
12	0.80 (0.57, 0.95)	0.21 (0.05, 0.45)
All months	0.60 (0.38, 0.81)	0.18 (0.04, 0.40)

Table for main text. Secondary analyses 3 and 4. Estimated median (95% CI) Community-based:Team-based temporal and spatial inter-village variance ratio. Population: all dogs with primary outcome data.

Analysis	Community-based:team-based ratio (95% CI)
Secondary 3 (variation over time)	0.67 (0.56, 0.83)
Secondary 4 (variation over space)	2.16 (0.05, 16.06)

Table for main text. Secondary analysis 5. Estimate (95% CI) V2:V1 ratio of Community-based:Team-based odds ratios. This ratio represents the interaction between trial arm and visit; the p-value tests the null hypothesis of equal intervention effect between visit 1 and visit 2. This interaction effect did not differ between the three years (trial arm × visit × year interaction p-value = 0.256). Population: all dogs with primary outcome data.

InteractionOR	P-value
1.58 (1.30, 1.92)	<0.001

Table 4. Secondary analysis 6. Estimate (95% CI) of V2:V1 odds ratios and of minimum coverage at visit 1 required for coverage at visit 2 to exceed 40%, by trial arm. V2:V1 odds ratios differed significantly between trial arms (interaction P

Trial_Arm	Visit effect OR (95% CI)	P-value	Start-of-year coverage for end-of-year coverage ≥ threshold
Team-based	0.60 (0.50, 0.72)	<0.001	0.61 (0.53, 0.68)
Community-based	0.94 (0.79, 1.13)	0.533	0.42 (0.35, 0.50)

Table S1. Estimates of fixed effects (log odds and log odds ratios) and random effects (variances) from the GLMMs fitted for the primary analysis. Numbers of observations, number of each random effect level, and marginal and conditional R^2 are also presented. Population: all dogs with primary outcome data.

	Primary analysis null hypothesis model				*Primary analysis alternative hypothesis model*			
Predictors	*Log-Odds*	*std. Error*	*CI*	*p*	*Log-Odds*	*std. Error*	*CI*	*p*
(Intercept)	-0.60	0.32	-1.23, 0.02	0.060	-1.48	0.37	-2.19, -0.76	<0.001
Visit: V2	-0.71	0.26	-1.22, -0.21	0.006	-1.11	0.32	-1.73, -0.49	<0.001
Year: Y2	0.16	0.26	-0.35, 0.66	0.542	0.65	0.31	0.04, 1.25	0.036
Year: Y3	0.90	0.26	0.40, 1.41	<0.001	1.11	0.31	0.51, 1.71	<0.001
VisitV2:YearY2	0.66	0.36	-0.05, 1.38	0.068	0.36	0.44	-0.50, 1.22	0.410
VisitV2:YearY3	-0.24	0.36	-0.95, 0.48	0.514	-0.14	0.44	-1.00, 0.72	0.751
Trial_Arm: Community-based					1.73	0.31	1.11, 2.35	<0.001
VisitV2:Trial_ArmCommunity-based					0.79	0.35	0.10, 1.49	0.026
YearY2:Trial_ArmCommunity-based					-0.93	0.34	-1.61, -0.26	0.007
YearY3:Trial_ArmCommunity-based					-0.37	0.34	-1.04, 0.30	0.281
VisitV2:YearY2:Trial_ArmCommunity-based					0.56	0.49	-0.40, 1.52	0.256
VisitV2:YearY3:Trial_ArmCommunity-based					-0.21	0.49	-1.17, 0.75	0.669
Random Effects								
σ^2		3.29				3.29		
τ_{00}		3.68 $_{\text{Household.file.id}}$				3.81 $_{\text{Household.file.id}}$		
		1.04 $_{\text{sub_village.rnd}}$				1.09 $_{\text{sub_village.rnd}}$		
		2.30 $_{\text{sub_village}}$				2.36 $_{\text{sub_village}}$		
		0.47 $_{\text{ward.rnd}}$				0.35 $_{\text{ward.rnd}}$		
		1.25 $_{\text{ward}}$				0.53 $_{\text{ward}}$		
		0.15 $_{\text{district.rnd}}$				0.16 $_{\text{district.rnd}}$		
		0.46 $_{\text{district}}$				0.58 $_{\text{district}}$		
N		12999 $_{\text{Household.file.id}}$				12999 $_{\text{Household.file.id}}$		
		2376 $_{\text{sub_village.rnd}}$				2376 $_{\text{sub_village.rnd}}$		
		659 $_{\text{sub_village}}$				659 $_{\text{sub_village}}$		
		663 $_{\text{ward.rnd}}$				663 $_{\text{ward.rnd}}$		
		112 $_{\text{ward}}$				112 $_{\text{ward}}$		
		54 $_{\text{district.rnd}}$				54 $_{\text{district.rnd}}$		
		9 $_{\text{district}}$				9 $_{\text{district}}$		
Observations		18358				18358		
Marginal R^2 / Conditional R^2		0.017 / 0.744				0.079 / 0.751		

Table S2. Characteristics of surveyed dogs, overall and by trial arm. Population: all dogs.

		Total (n=25677)	Trial Arm	
			Team-based (N=12670)	Community-based (N=13007)
Age (years)	N_{OBS} ($N_{MISSING}$)	25677 (0)	12670 (0)	13007 (0)
	Mean (SD)	3.1 (2.1)	3.1 (2.2)	3.0 (2.1)
	Median (IQR)	3.0 (1.0, 4.0)	3.0 (1.0, 4.0)	3.0 (1.0, 4.0)
	[Range]	[0.5, 16.0]	[0.5, 16.0]	[0.5, 16.0]
Sex	N_{OBS} ($N_{MISSING}$)	25677 (0)	12670 (0)	13007 (0)
	Female	8527 (33%)	4003 (32%)	4524 (35%)
	Male	17150 (67%)	8667 (68%)	8483 (65%)
Year-visit	N_{OBS} ($N_{MISSING}$)	25677 (0)	12670 (0)	13007 (0)
	Y1-V1	4423 (17%)	2227 (18%)	2196 (17%)
	Y1-V2	4060 (16%)	1988 (16%)	2072 (16%)
	Y2-V1	4297 (17%)	2078 (16%)	2219 (17%)
	Y2-V2	4503 (18%)	2224 (18%)	2279 (18%)
	Y3-V1	4222 (16%)	2091 (17%)	2131 (16%)
	Y3-V2	4172 (16%)	2062 (16%)	2110 (16%)
Primary coverage outcome	N_{OBS} ($N_{MISSING}$)	18358 (7319)	9214 (3456)	9144 (3863)
	Not vaccinated	9325 (51%)	5177 (56%)	4148 (45%)
	Vaccinated	9033 (49%)	4037 (44%)	4996 (55%)
Secondary coverage outcome	N_{OBS} ($N_{MISSING}$)	25677 (0)	12670 (0)	13007 (0)
	Not vaccinated	16644 (65%)	8633 (68%)	8011 (62%)
	Vaccinated	9033 (35%)	4037 (32%)	4996 (38%)

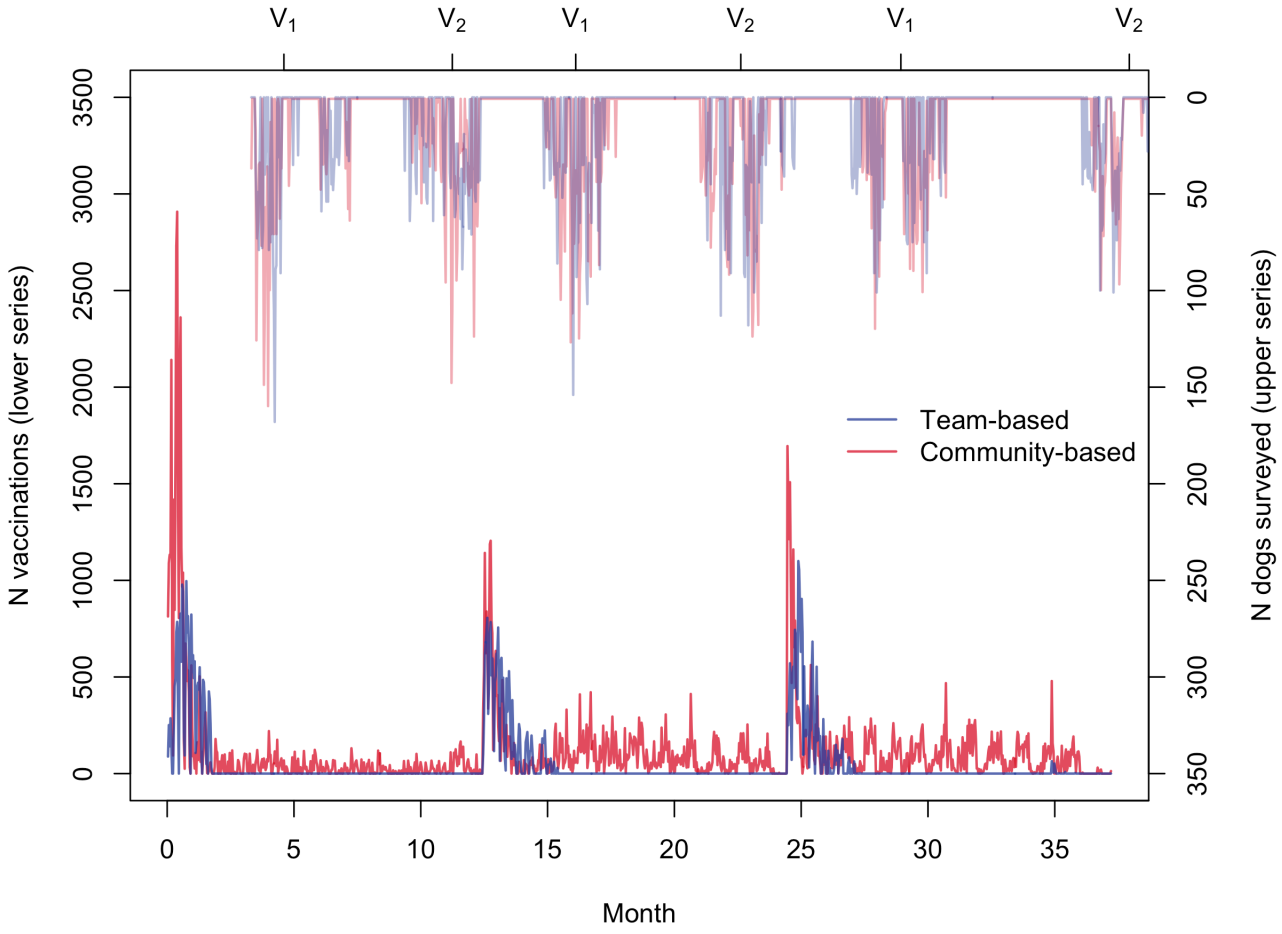

Figure 2. Number of rabies vaccinations performed (lower series; total = 186931) and number of dogs surveyed (upper series; total = 25677) per day, by trial arm.

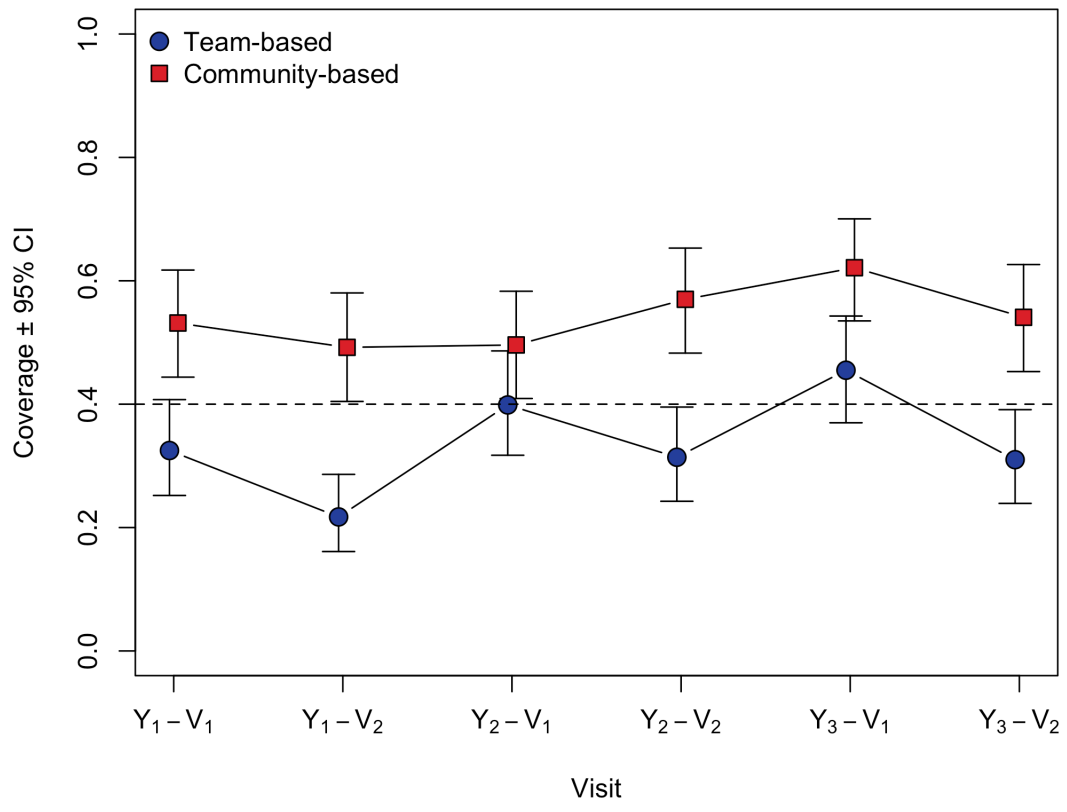

Figure 3. Estimated coverage ± 95% confidence limits at each survey time point, by trial arm. Population: all dogs with primary outcome data.

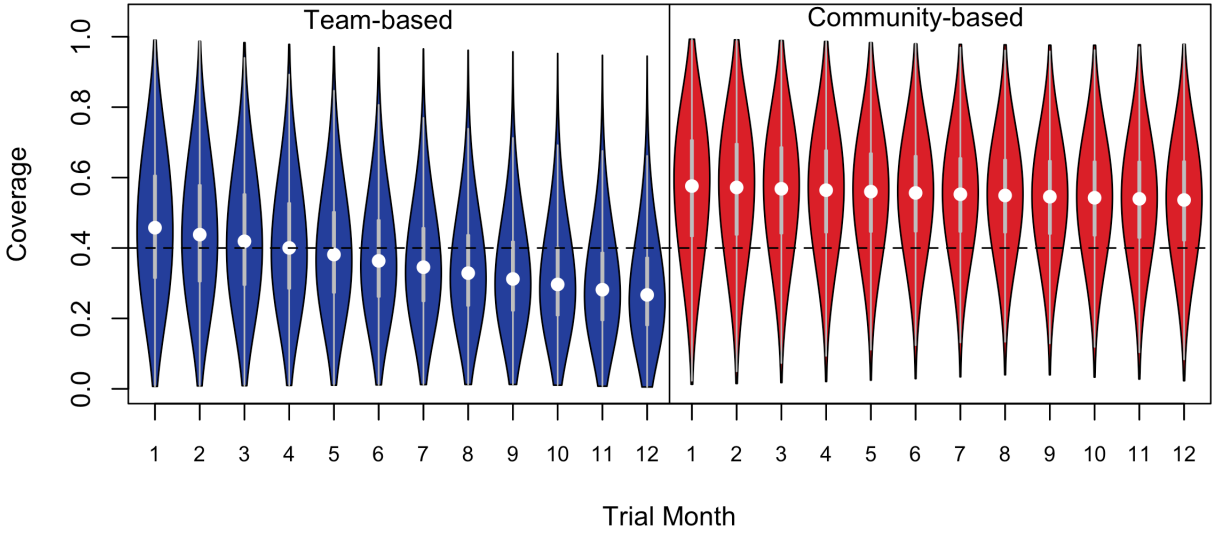

Figure 4. Estimated probability distribution of coverage by trial month for each arm. The coverage target of 40% is shown by a dashed line. White circles indicate medians and quartiles and range are shown by box-and-whisker plots.

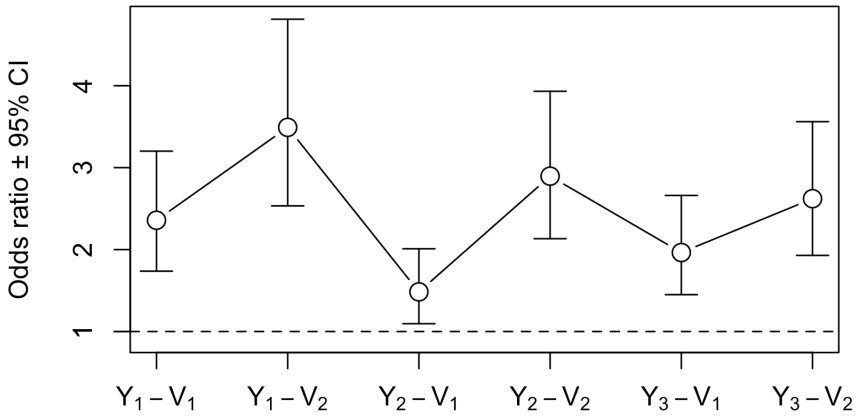

Figure 5. Estimated intervention effect (Community-based:Team-based) odds ratios ± 95% confidence limits at each survey time point. Population: all dogs with primary outcome data.

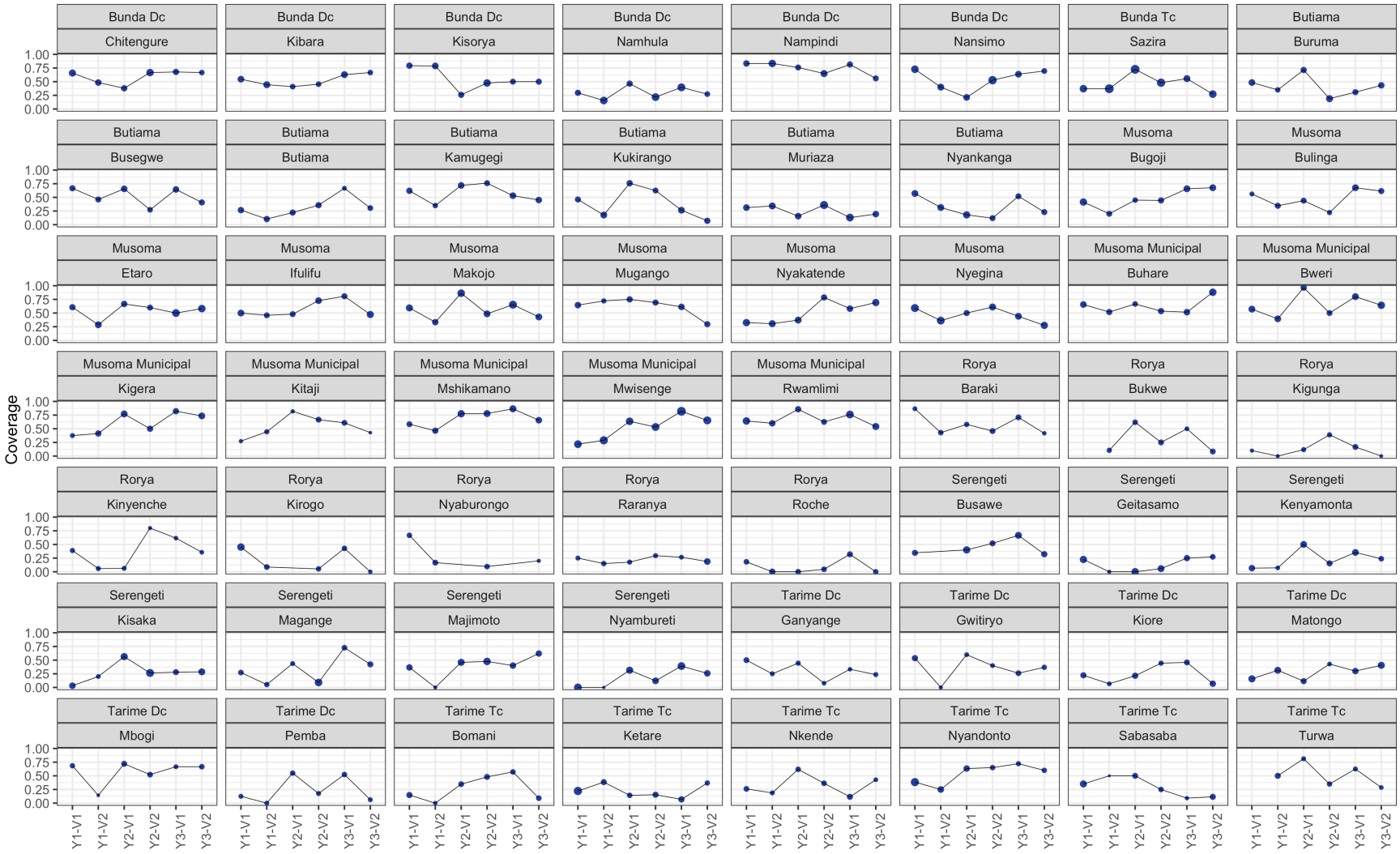

Figure S1. Coverage by ward and survey time point in the Team-based arm.

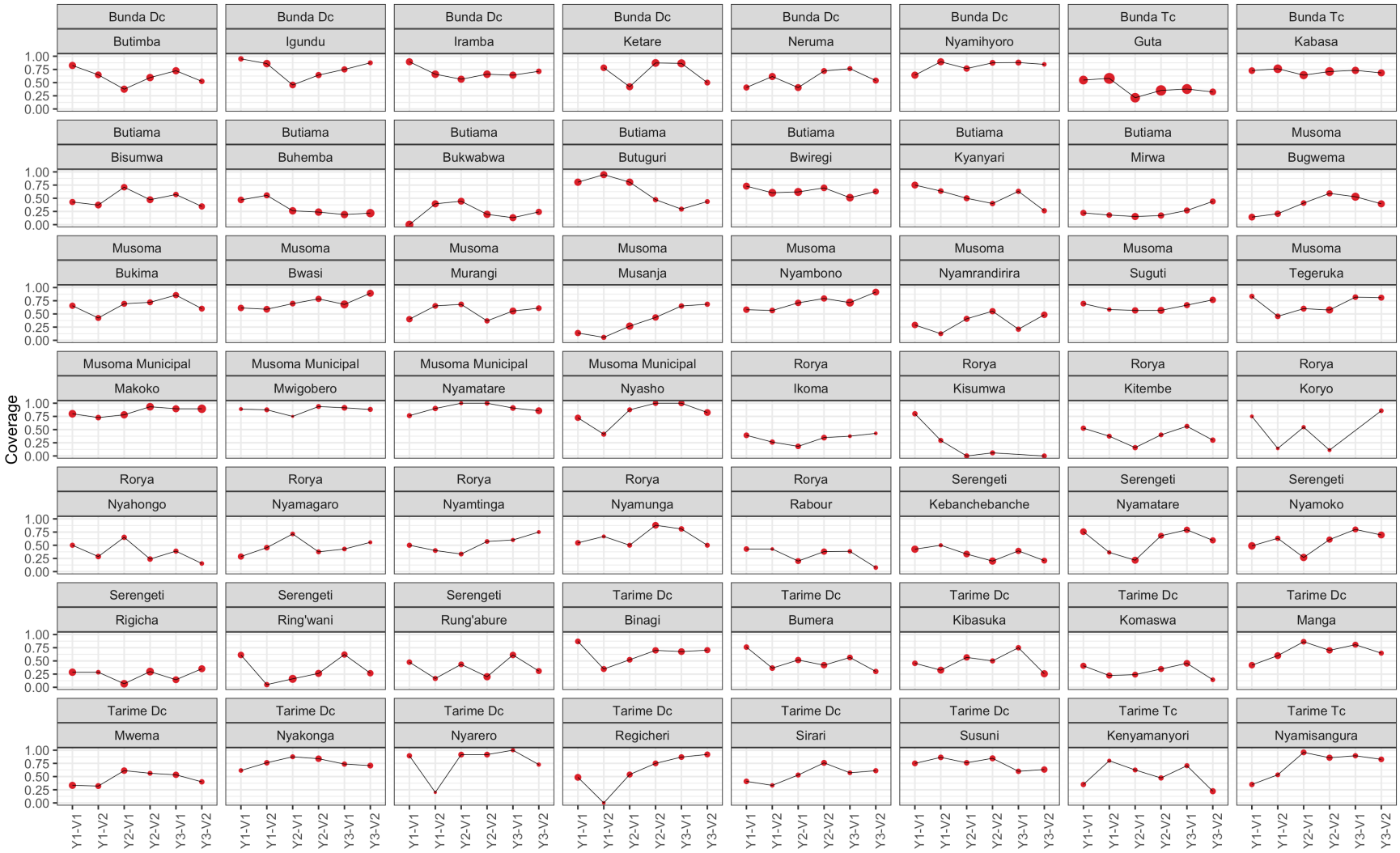

Figure S2. Coverage by ward and survey time point in the Community-based arm.